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Wood Hybrids: multi-species spatial timber structures

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Abstract

Hybrid wood species design is a proposed concept that integrates underutilised low-density, fast-growing softwood and reclaimed hardwood. This concept takes advantage of both types of wood: hardwood provides strength and stiffness and softwood provides volume and reduced weight. It also emphasises the importance of localised material sourcing and availability-based design thinking. In this study, reclaimed hardwood focuses on the materials retrieved from building demolition processes. To test and demonstrate the hybrid wood species design concept for architectural and structural assemblies, we present *Wood Hybrids*, a pavilion structure that assembles 28 linear elements to define a spatial enclosure. The pavilion structure was driven by a material availability-based concept and demonstrated how multiple elements can be joined at one node through innovative wooden connections, which specifically included the use of locally salvaged wooden window frames. A critical loading case was analysed, followed by each element designed using a developed design workflow. The pavilion highlighted how wood hybridisation could become not only a means of improving structural efficiency and material diversity, but also a concept and application that explicitly expresses the logic and arts of a structure.

Keywords: hybrid wood species design, design for availability, spatial timber joints, fast-growing softwood, reclaimed hardwood

1. Motivation

The German architect Gottfried Semper once said "*let the material be true to itself*" [1], which emphasises the nature of statical laws of the material. Building on it, we propose "*let the available material be true to itself*", meaning that beyond the statical laws of the material, the history, status, and possible future transformation of the material are also respected and integrated in the analysis and design process. *Wood Hybrid* pavilion was therefore conceptualised and designed in line with this concept, where we appreciated the quality of available wood materials, either new or used, emphasising the importance of localised material sourcing and availability-based design thinking. Through simple material processing, the materials can find their purposes and express their structural and architectural qualities in a new context and at the same time maintaining their original features and characteristics [2].

In this paper, we advance hybrid wood species design for structural beam-type elements and spatial wooden joints. A specific design process in line with material availability-based design thinking was developed that started from both initial material inputs and a predefined geometric form, exploring suitable material hybridisation for the design of individual elements and joints. Aspects including geometric and mechanical properties of different wood species were integrated into the workflow. Design objectives, such as maximising material utilisation, were defined as quantitative output targets, while joinery articulation and visual characteristics of structural members were qualitatively evaluated.

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2. Design for material availability

Different from the conventional design approach that does not or only considers availability issues at the latest stages, availability-based design emphasises early involvements of material availability in design process [3]. This is particularly relevant in the context of circular construction, where reused or upcycled materials often have non-uniform dimensions and varying properties, resulting in design challenges when using them. Design for material availability, therefore, attempts to find the best match between material availability and design solutions at early stages, which can significantly contribute to a holistic design outcome [4].

In the history of construction, it was common practice for master builders to search for locally accessible and affordable materials when planning and designing buildings [5]. For example, in timber construction, crooked trees that are not used for structural applications nowadays were particularly sought after and used for specific parts of roof trusses [6]. This demonstrates how material availability can influence a structure's core characteristics, provided the properties of the available materials are respected and preserved. However, this possibility largely disappears if a design only considers how to fit materials into an intended geometry.

Consequently, the proposed availability-based approach to structural design starts from associating general geometry with material availability. By analysing available materials and drawing insights from the analysis, strategies for forming individual structural elements and joining them to create large assemblies are sought to match the intended general geometry, through continuous interaction between the four main pillars in the resulting loop, as illustrated in Figure 1 (left).

3. Wood Hybrids pavilion

To demonstrate the proposed material availability-based approach,we present the Wood Hybrids pavilion, which was built on the framework in Figure 1 (left). The pavilion design started from both material availability and general geometry, followed by explorations on individual elements and spatial wooden joints. These aspects were linked by hybrid wood species design, originated from explorations on boat-building and furniture-making [7]. The concept of wood hybridisation was applied in two senses. One was hybrid linear elements, where the individual elements with hybrid cross-sections using different wood species were explored, inspired by the experimental investigations performed in [8]. The other was hybrid spatial wooden joints, where salvaged wooden window frames were innovatively utilised for connecting spatial timber elements. The conceptual design process of the pavilion is illustrated in Figure 1 (right).

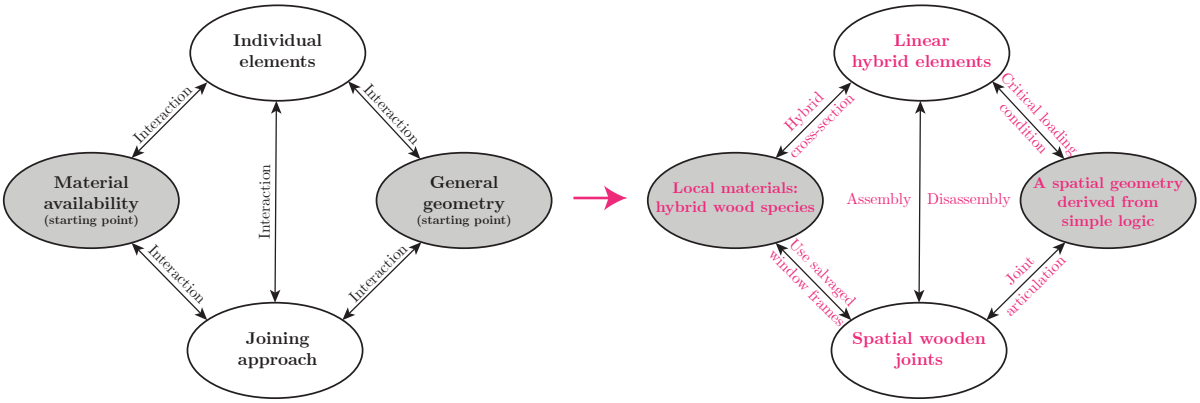


Figure 1: Design for material availability for structural design: conceptual design process (left) and its application in Wood Hybrids pavilion (right).

3.1. Available materials

The materials used for the pavilion were mainly fast-growing, low-density softwood (grand fir with an average density of around 390 kg/m^3), reclaimed hardwood (oak with an average density of around 700 kg/m^3), and salvaged wooden window frames (mostly pine) (Figure 2). All the materials were locally sourced: the grand fir woods were harvested from local planted forests as planks with a length of 2 m (Figure 2a), which were not originally used for structural purposes; the reclaimed oak woods were sourced from disassembled floor panels nearby with specific cross-sections of 20 mm × 65 mm or 20 mm × 100 mm and varied lengths (Figure 2b); and the wooden window frames were salvaged from a local recycling centre with various sizes (Figure 2c). Note that the salvaged window frames were carefully selected according to their manufacturing dates to avoid the potential presence of polychlorinated biphenyls between the window panes.

3.2. General geometry

The general geometry of the pavilion was formed by a simple logic, derived from a basic 2 m × 2 m square at the ground level, moving to two height levels at 2 m and 4 m, as seen in Figure 3. The final position of the square at 2 m height was defined by a 1 m horizontal shift and a 45° in-plane rotation, while the square at 4 m height remained unchanged. The corners of the three squares were connected in a specific way: each corner of the middle square was connected to two adjacent corners of the other two squares, forming, in total, 16 elements between the squares (diagonal elements). This defined a spatial structure consisting of 28 elements, resulting in a specific structural rhythm while maintaining geometric simplicity.



(a) Grand fir wood planks (b) Reclaimed oak (c) Waste window frames

Figure 2: Available materials for the pavilion.

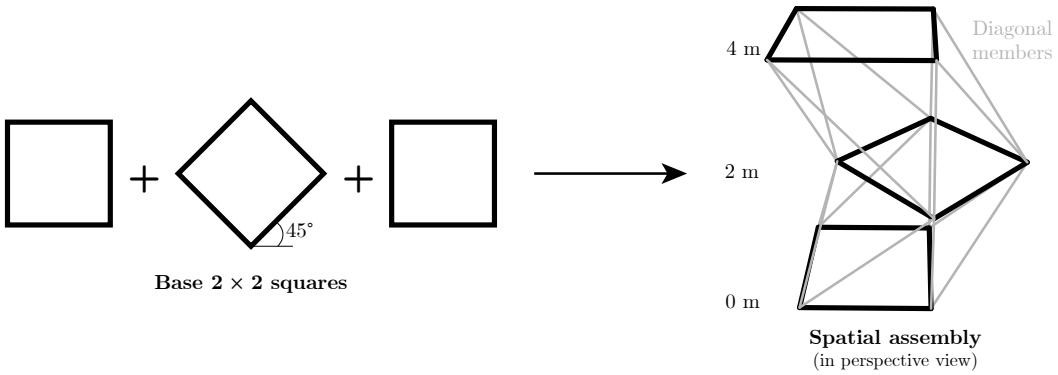


Figure 3: The deriving logic of the pavilion geometry.

3.3. Structural design of individual elements

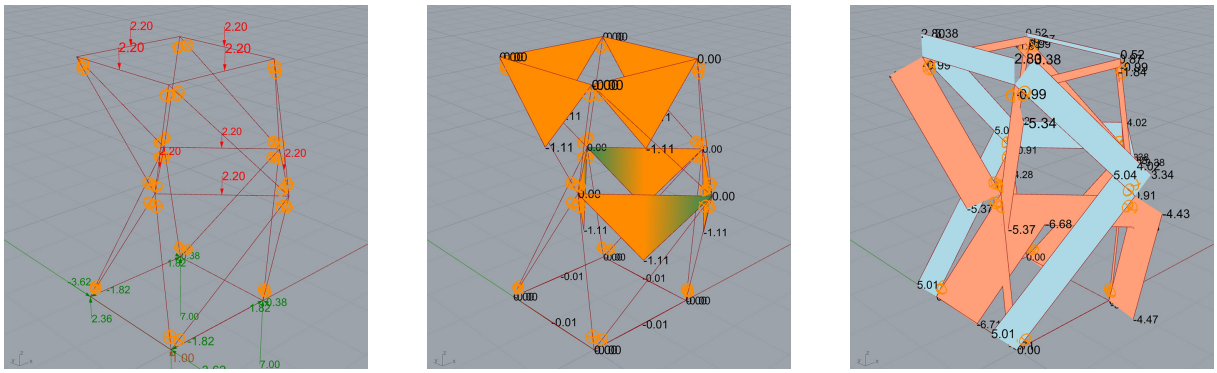
The linear elements of the pavilion were designed based on the stress distribution of each element in the structure. Under a specific loading condition, the structure was statically analysed. According to the results of structural analysis, hybrid wood species design concept was applied, where a series of designs with different cross-sectional compositions were generated using a developed design workflow. Through using high-density reclaimed hardwood for high stress area and low-density fast-growing softwood for the rest, the appearance of each element explicitly expressed the load concentration within the element. Here, the hybridisation was realised through glueing for this pavilion.

3.3.1. Structural analysis of entire structure

The structure was analysed under a potential critical loading condition considering future installation of floor elements, dominated by a vertical pointed load (2.2 kN) at each side of the square at height level 2 m and 4 m (Figure 4a). Assumptions were made for the structural analysis. The corners of the squares were considered as rigid in the horizontal plane, based on the fact that reclaimed window frames with their ready-made corner joints could provide in-plane stiffness. All vertical connections (diagonal to square) were considered as hinge, meaning that the vertical elements were only subject to axial loads. The structural analysis was performed using Karamba3D [9], a finite element plug-in in the Rhino Grasshopper environment [10]. The diagrams of bending moment and normal forces are shown in Figure 4b and 4c.

3.3.2. Design of individual linear elements

The results of structural analysis were subsequently used for designing individual elements. The workflow was built upon the performed experimental investigations on hybrid wood species design [8], which demonstrated that a hybrid cross-section could be transformed to a mono-material cross-section according to elastic modulus. The designs of individual elements followed the material availability and its associated limits, i.e., the maximum length of grand fir planks (2 m) and the original thickness of oak (20 mm). Three types of cross-sections, including one pure grand fir and two hybrid types, were considered in the design of elements (Figure 5). A constant cross-section of $b \times h = 60$ mm $\times$ 80 mm was applied to all elements, estimated by analysing the structure using grand fir timber with material properties equivalent to C16 class. Each element was then designed based on a reference minimum design through assigning the three cross-sections according to its stress distribution, where other aspects, such as low processing and aesthetic quality, were also considered for the final design decision-making (Figure 5).



(a) Loads and reactions (b) Bending moment diagram (c) Normal force diagram

Figure 4: Loading condition and internal forces of the pavilion structure.

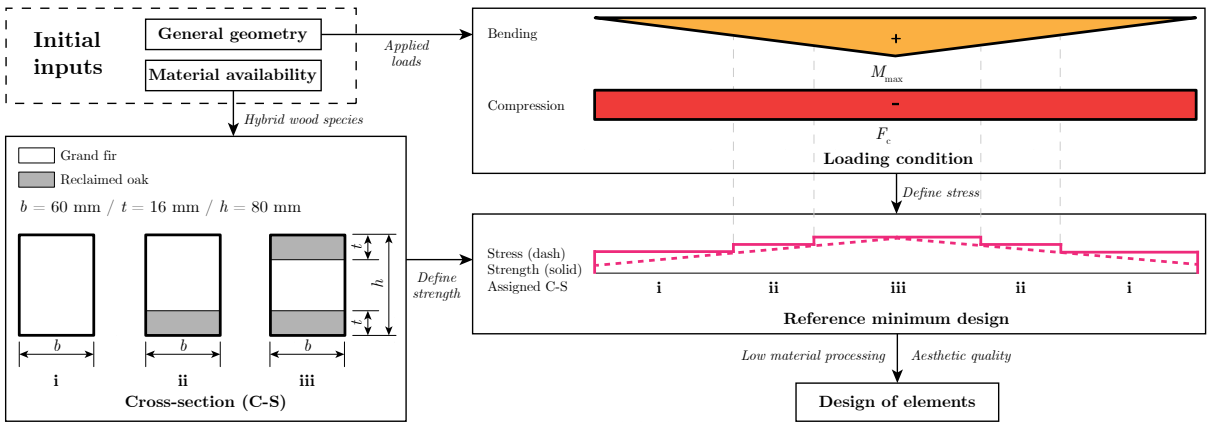


Figure 5: Conceptual design workflow of an element subjected to bending and compression, where $M_{\max}$ represents the maximum bending moment and F_c represents the compression force. To be mentioned that (i) as the influence of shear is small in the present case, shear diagram is therefore not included here, and (ii) lateral buckling is prevented through bracing when it is needed.

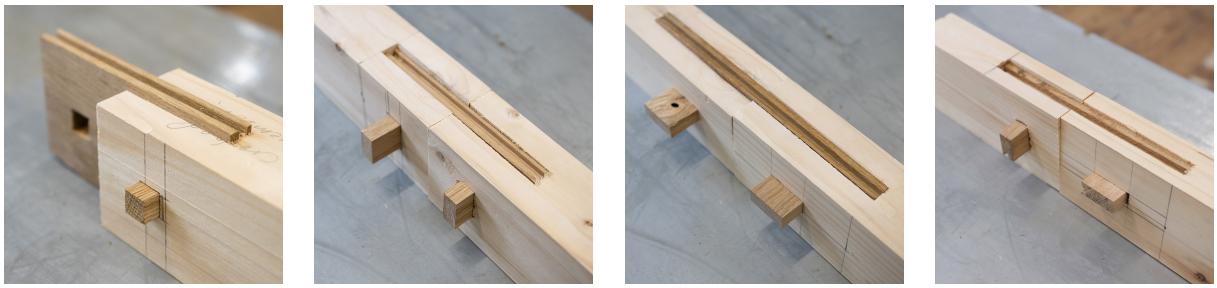
3.4. Wooden joints for spatial timber assembly

There were two types of joints developed for the pavilion. One was linear joint aimed to assemble two linear parts to form a diagonal element that was only subjected to axial load. This was primarily driven by the desire to create joints for assembling short elements and the consideration of material availability. Another type was spatial joint, realised by a specific use of corner joints of salvaged window frames collected from a local recycling centre. As the undamaged corner joints still held sufficient structural quality, they could be directly used for structural applications instead of making new joints. The two types of joint were developed through a process of design and prototyping involving several iterations. Following is the detailed joint-developing process.

3.4.1. Linear joints

The linear elements subjected to pure tension or compression, therefore, were mainly used for keeping two linear parts in place in the case of compression element and for providing tensile resistance in the case of tension element. A dowel-type connection was applied based on the available materials, i.e., reclaimed oak was used as both dowels and inserted plate to connect the two segments of a beam. The reclaimed oak was low-processed from the original floor panels with a thickness of 20 mm, i.e., planed 2 mm from each flat edge down to 16 mm, and then cut into designed geometries. For the size and spacing of dowel, the spacing requirement for dowel-type fastener in EN-1995 [11] was considered.

As an initial prototype L1, a square dowel with a cross-section of 16 mm × 16 mm (Figure 6a) was produced. The use of a square dowel rather than a round one was based on the consideration of low material processing, as the square dowel could be easily achieved from simple cutting. However, the square dowel provided limited rotational resistance, leading to further consideration of using rectangular dowels. Consequently, rectangular dowels with a cross-section of 16 mm × 40 mm were produced. The long side of the dowel was either perpendicular (L2) or parallel to the grain (L3), as seen in Figure 6b and 6c. The longer dowels provided better rotational resistance and tensile strength. However, the prototype L2 resulted in a small spacing distance to the edge perpendicular to the grain, while the prototype L3 led to an increased insertion (cutting-through) length into the softwood. As a solution, one perpendicular and one parallel dowel (L4) were used for the linear joints to provide rotational resistance, which also created a unique expression of the linear joints, as seen in Figure 6d.



(a) Prototype L1 (b) Prototype L2 (c) Prototype L3 (d) Prototype L4

Figure 6: Exploratory prototypes for the linear joints with different arrangements of dowels.

3.4.2. Spatial joints

Grand fir, reclaimed oak and salvaged wooden window frames were used for making the spatial joints. A spatial timber joint could be formed by connecting reclaimed oak pieces to several structural elements from various orientations through a layered arrangement of corner joints from salvaged window frames that created gaps in between. To explore an elegant and structurally efficient joining approach, prototypes were produced and tested, taking into account easy fabrication and a simple assembly/disassembly process. The initial prototypes partly used glued L-shaped joints made from two pieces of softwood to mimic the corner joints of window frames.

In the first prototype S1 (Figure 7a), the softwood directly contacted the corner joints so that the axial load from the diagonal elements was directly transferred to the corner joints, while the hardwood parts were used only for keeping the parts in place. However, in this case the high strength of hardwood was not fully utilised, and the softwood was also weakened by cutting.

In the second prototypes S2 (Figure 7b), reclaimed oak piece was inserted into the gap between two corner joints, and another oak piece (connected to diagonal elements) was inserted into the first oak piece through an angle-cut and locked by dowels in two directions. In this case, the hardwood was the main contributor to load transfer, but its axial capacity was not fully utilised.

In the third prototype S3 (Figure 7c), two double-layer reclaimed oak pieces were used. One was inserted into the gap between the window frames and the other was used to adjust the orientation of the diagonal elements. The joint was formed by cutting the two oak pieces at precise, matching angles. The pieces were glued together to form a wooden connector, which was slid into the gap between the window frames and secured with dowels.



(a) Prototype S1 (b) Prototype S2 (c) Prototype S3

Figure 7: Exploratory prototypes for the spatial joints with different configurations.

3.5. Assembly of the pavilion

The design of the pavilion was decided based on the explorations of elements and joints, their interactions with material availability and general geometry, as well as considerations of low processing, simple assembly and aesthetics. The final design of the pavilion is shown in Figure 8.

Three different types of horizontal elements were designed: element type-1 using cross-section type-i, element type-2 using cross-section-type ii, and element type-3 using a combined cross-section type-ii and iii. In the element type-3, cross-section type-iii was located in the middle of element with a length of 400 mm. The diagonal elements were all in type-i cross-sections either with or without linear joints. Individual elements were fully prefabricated, including the assembly of linear joints, which was based on the logic developed during the joint explorations.

The individual elements were subsequently assembled using the developed spatial joints, which were designed to accommodate the pavilion's elements in their various orientations. A total of 12 spatial joints were made: eight at the top and bottom levels for connecting four elements at one node, and four at the middle level for connecting six elements at one node. The salvaged window frames and reclaimed oak were first assembled as a spatial connector using the similar concept of the spatial joint prototype S3, and then they were connected to the horizontal and diagonal elements via dowel connections.

The pavilion can be assembled from the bottom up and fully disassembled in reverse.

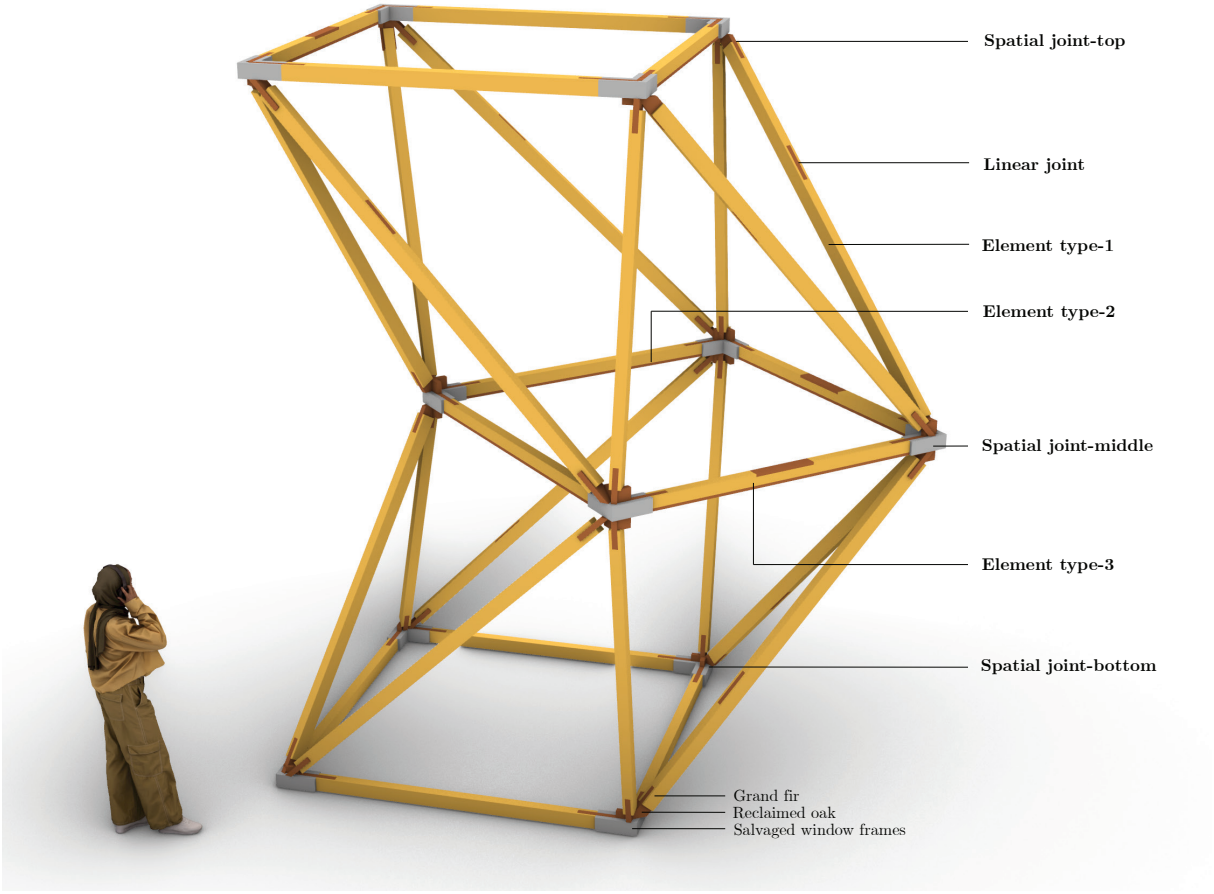


Figure 8: Rendering of the assembled pavilion (the joint detail may differ from the final construction).

4. Discussions on interactions

The interactions between the four main pillars included in the framework of design for material availability are extensively discussed here. An overview of the interactions is illustrated in Figure 9.

Available materials – Individual elements: The availability of materials shaped the design of the individual elements. The hybrid cross-sections were defined according to the thicknesses of the grand fir planks and reclaimed oak panels, taking into account the reduction in thickness resulting from planing down both surfaces. The length of each element followed that of the grand fir plank, ensuring that the full length of the planks could be utilised in most cases. The design of the individual elements also influenced the selection of the reclaimed oak panels. Due to the various lengths of the panels, the most suitable ones in terms of minimising waste were selected for element fabrication.

Available materials – Joining approach: The availability of materials defined the joining approach in three ways. Firstly, the requirement for linear joints in the diagonal elements was a consequence of the material limitation and the motivation of using short elements. Secondly, the square or rectangular shape of the dowel connection was chosen because it required minimal processing of the original panels – only simple cutting was needed. Thirdly, the salvaged window frames were used innovatively by maintaining their original corner joints for structural purposes. In reverse, the development of joints defined specific material processing approaches, e.g., the square and rectangular oak dowels could be easily made through cutting.

General geometry – Individual elements: The predefined geometry resulted in specific loading conditions for the elements in the event of a critical loading scenario. The design of each element was based on the reference minimum design using hybrid wood species, resulting in a unique appearance that aligned with the stress distribution. The design of the elements could possibly be optimised to, for example, minimise material waste or use of virgin wood.

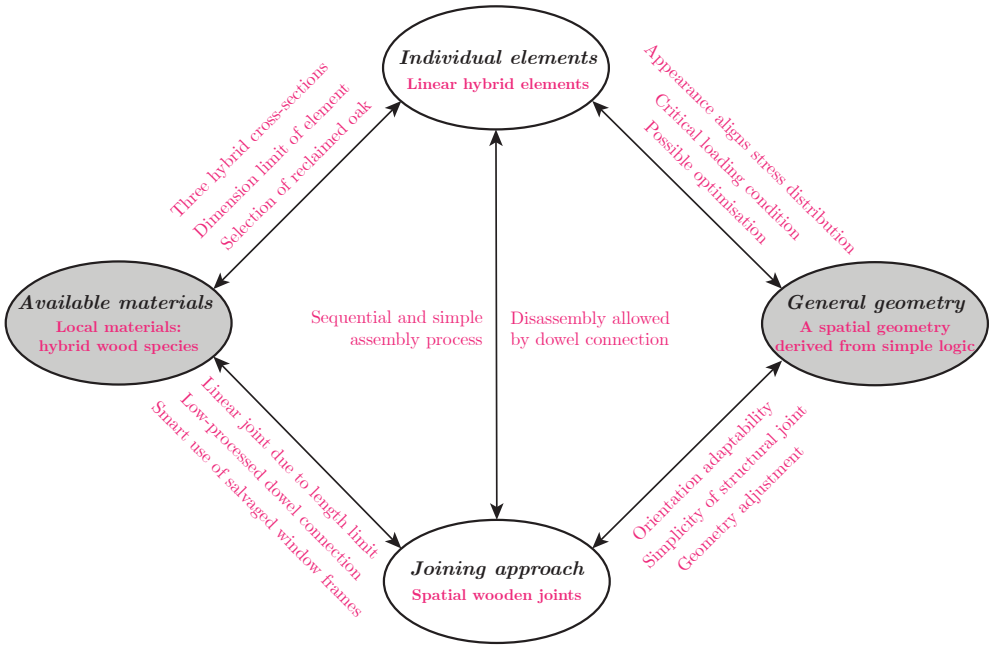


Figure 9: Detailed interactions between the four main pillars in the framework of design for material availability in the pavilion design.

General geometry – Joining approach: The general geometry defined complex spatial joints that needed to bring together six elements with different orientations. This was achieved through a specific design of the joints, which incorporated short reclaimed oak pieces to alter the orientation. The spatial joints also demonstrated simplicity and clear structural logic: the salvaged window frames provided horizontal in-plane stiffness, while the reclaimed oak connected the elements and created the intersections and load transfer. Due to this joining approach, the general geometry had to be adjusted to leave enough space to allow the elements to be joined together at one node. This means that the central line of each element may differ slightly from the geometry originally derived in Figure 3.

Individual elements – Joining approach: Simple assembly and disassembly of the individual elements were considered throughout the development of joints. The individual elements were joined using a sequential assembly logic that could also be reversed for disassembly, thanks to the dowel connections. Furthermore, the cross-sectional dimensions of the elements were closely linked to the size and arrangement of the dowels used for the joints.

5. Conclusion

The Wood Hybrids pavilion showcased the proposed hybrid wood species design concept through using local fast-growing, low-density softwood, reclaimed hardwood, and salvaged window frames, emphasising the importance of localised material sourcing and availability-based design thinking. A design framework for material availability was suggested, supported by four main pillars: material availability, general geometry, individual elements and joining approach. Given the predefined material availability and general geometry, the pavilion was designed using a workflow developed for hybrid linear elements and an exploratory approach to hybrid wood spatial joints. The final design was determined through the active interaction of these four pillars with considerations of low material processing, simple assembly/disassembly and aesthetic quality.

The main contributions of the demonstration pavilion can be concluded as:

- The pavilion exemplifies the hybrid wood species design concept that prioritises material availability as a possible design approach to tackle the challenges of responsibly using and reusing wood-based materials.
- The pavilion demonstrates the potential to add not only structural but also aesthetic values to reclaimed materials and components through showcasing their unique characteristics in structural applications.
- The linear and spatial joints developed in this study show how tectonic value can be introduced through joint detailing and making. This reveals the meaning of each wood species within the hybrid joint system, considering their status and histories and expressing them through the logic of the overall structural composition.

Based on this study, future work will focus on structural testing and evaluation of the developed spatial joints. Research interests will also focus on investigating plate-type elements using the hybrid wood species design concept, as well as their structural, thermal and acoustic properties.

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